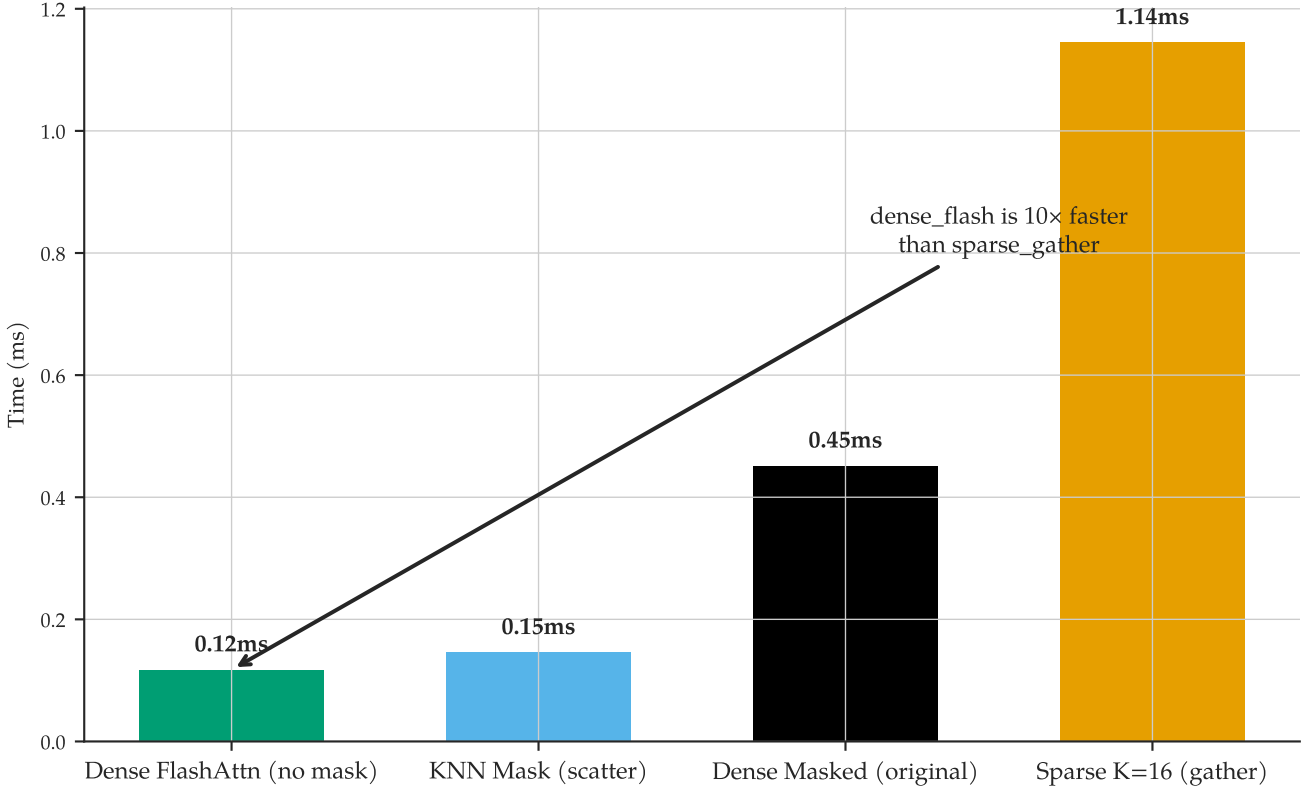
Attention Mechanism Comparison at N=256
Time (ms)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1.14ms
dense_flash is 10× faster
than sparse_gather
0.45ms
0.15ms
0.12ms
Dense FlashAttn (no mask)
KNN Mask (scatter)
Dense Masked (original)
Sparse K=16 (gather)